

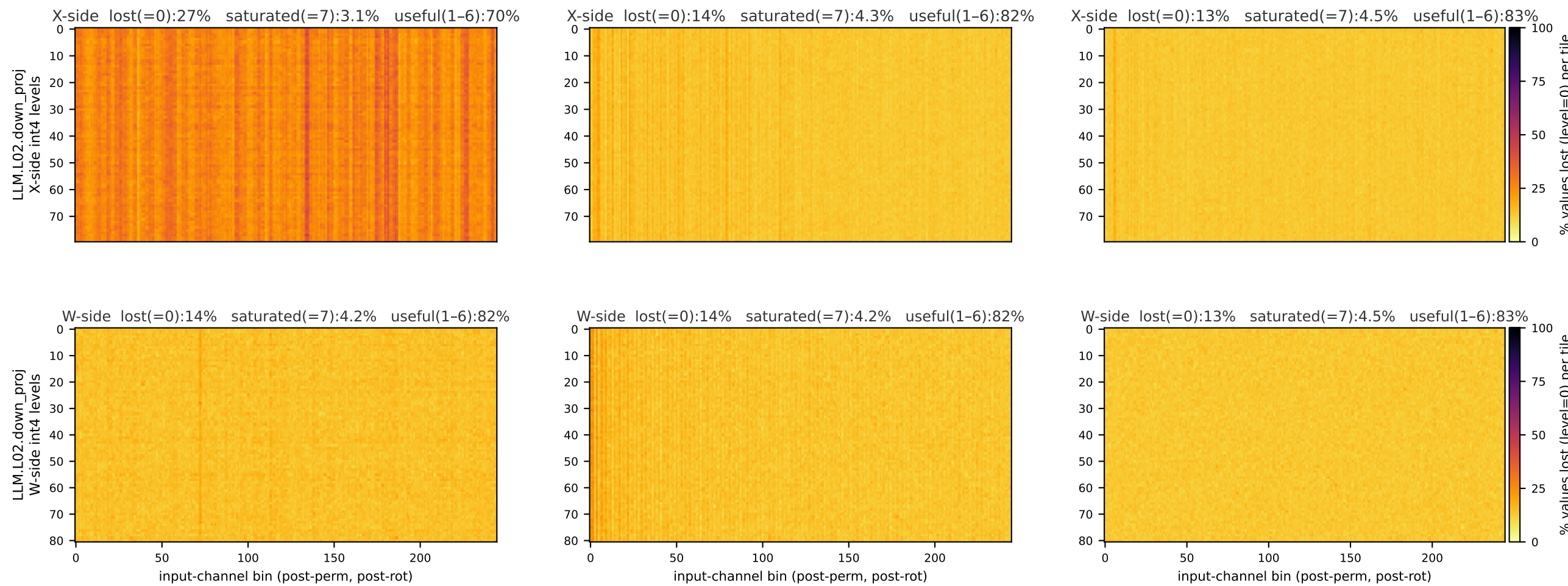
gr00t-N1.5 — DuQuant SVD vs SVD-Hadamard (A2-lite) — int4-utilization heatmap
Color = % of values 'lost' (rounded to int4 level 0) per tile, after per-block MSE-optimal scaling.
Dark stripes = outlier blocks throwing away most int4 levels; uniform yellow = bits well-spent.

raw (no rotation)

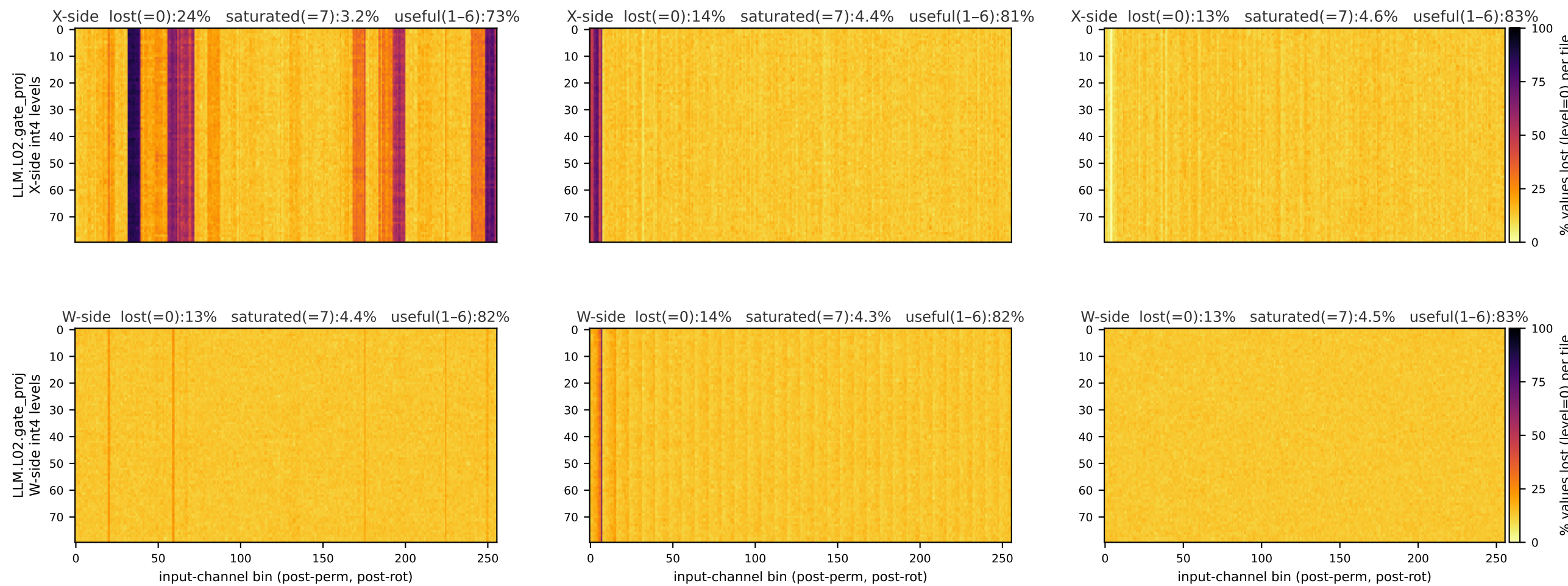
DuQuant SVD

SVD-Hadamard (A2-lite)

LLM.L02.down_proj — structural pathology (raw amax-ratio 48000× ; SVD-H still 11400×)



LLM.L02.gate_proj — typical MLP layer (SVD-only leaves amax-ratio 56×, SVD-H drops it to 14×)



raw (no rotation) — level distribution (LLM.L02.gate_proj)

DuQuant SVD — level distribution (LLM.L02.gate_proj)

SVD-Hadamard (A2-lite) — level distribution (LLM.L02.gate_proj)

